

When Local Expert Geometry Meets Discrete Routing: Functional Response and Routing Transitions in Sparse Mixture-of-Experts Models

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Abstract

Sparse mixture-of-experts (MoE) models combine continuous expert computation with discrete top- k routing. Can the distributional effect of jointly changing expert contributions be predicted from a local expert geometry, and when does this approximation fail? We define an expert functional response matrix $F = \mathbb{E}_{x,y \sim p_0} [s(x,y)s(x,y)^\top]$, where each coordinate of s is the derivative of a next-token log probability with respect to one post-gating expert contribution. One backward pass recovers all 128 expert coordinates. On Qwen3-30B-A3B-Base and a fixed GSM8K protocol, the quadratic form $\frac{1}{2}\delta^\top F \delta$ predicts mean held-out KL for 40 local joint interventions with $R^2 = 0.837$ and Pearson $r = 0.915$. The diagonal baseline is statistically indistinguishable ($R^2 = 0.841$; paired bootstrap $\Delta R^2 = -0.004$, 95% CI $[-0.021, 0.029]$), so the data do not support an advantage from estimated off-diagonal interactions. At larger perturbations, only 17 of 1,920 intervention-prefix records change downstream routing, but their mean KL is 2.47×10^5 times the no-flip mean. A normalized perturbation-to-router-margin statistic separates flip from no-flip records with ROC-AUC 0.998. These results support a local response regime, identify routing transitions as a rare high-impact failure mode, and reveal substantial prefix-sampling instability that limits global geometry claims.

1 Introduction

Sparse MoE models scale capacity by sending each token to only a few experts (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Most MoE analyses focus on router probabilities, load balance, or activation counts. These describe which experts are selected, but not how changing an expert’s realized contribution changes the predictive distribution. That distinction matters when experts have redundant, synergistic, or opposing downstream effects.

We study a local functional geometry over expert contributions. At a chosen layer and token, we intervene after top- k selection and gate normalization, scaling expert i ’s realized contribution by $1 + \delta_i$. The derivative of a next-token log probability with respect to δ_i defines a functional score. Its second moment forms a positive semidefinite response matrix F , analogous to a Fisher matrix in expert-intervention space. Locally, $\frac{1}{2}\delta^\top F \delta$ predicts the KL divergence between baseline and intervened next-token distributions.

This local description coexists with a discontinuity. Routing at the intervened layer is fixed by construction, but the perturbed hidden state is propagated normally and later routers may change top- k sets. We therefore separate a smooth routing-stable regime from a routing-transition regime in which a perturbation crosses a boundary.

Our contributions are:

- An exact post-gating intervention whose full 128-dimensional response is obtained by one backward pass, with algebraic and finite-difference tests.

- Evidence that an $N = 512$ response estimate predicts unseen local joint interventions on held-out prefixes, but no evidence that full F improves over its diagonal.
- A router-margin analysis showing that rare downstream flips are sharply predictable and coincide with orders-of-magnitude larger KL effects.
- Negative stability results across more token samples, Rao–Blackwellization, Hutchinson probes, and shrinkage; an independently reserved confirmation set remains sealed.

2 Related work

Sparse mixture-of-experts models. Sparsely gated layers increase capacity without proportional per-token computation (Shazeer et al., 2017). GShard and Switch Transformers developed scalable routing systems (Lepikhin et al., 2021; Fedus et al., 2022), and later work studied stable transfer at scale (Zoph et al., 2022). Routing frequency, however, does not measure an expert’s functional output effect.

Local geometry and interventions. Fisher information describes local second-order distributional change under parameter perturbations (Amari, 1998). Our coordinates are not model parameters; they are temporary dimensionless scalars on post-gating expert contributions. The discrete top- k router creates piecewise-smooth regions separated by routing boundaries, whose effect we measure explicitly.

3 Expert functional response

3.1 Contribution-space intervention

For one MoE layer and token, let $h \in \mathbb{R}^d$ be the layer input, $f_i(h)$ expert i ’s output, and $g_i(h)$ its normalized top- k gate. The realized contribution and baseline output are

$$u_i = g_i(h)f_i(h), \quad z_0 = b + \sum_{i=1}^E u_i. \quad (1)$$

We introduce $\delta \in \mathbb{R}^E$ and compute

$$z_\delta = b + \sum_{i=1}^E (1 + \delta_i)u_i, \quad \Delta z = U\delta. \quad (2)$$

The intervention is after gate normalization. It changes no expert input or parameter, performs no second normalization, and preserves the target layer’s baseline expert identities. Inactive experts have zero local score. Downstream transformer and router layers execute naturally on the perturbed hidden state.

3.2 Response matrix and KL prediction

For prefix x , let $p_0(y | x)$ and $p_\delta(y | x)$ be baseline and intervened next-token distributions. Define

$$s_i(x, y) = \left. \frac{\partial \log p_\delta(y | x)}{\partial \delta_i} \right|_{\delta=0}, \quad F = \mathbb{E}_{x, y \sim p_0(\cdot | x)} [s(x, y)s(x, y)^\top]. \quad (3)$$

One backward pass through differentiable δ yields all E coordinates. We accumulate outer products in FP64:

$$\hat{F} = \frac{1}{N} \sum_{n=1}^N s_n s_n^\top. \quad (4)$$

Under regularity within a fixed-routing region,

$$\mathbb{E}_x D_{\text{KL}}(p_0 \| p_\delta) = \frac{1}{2} \delta^\top F \delta + o(\|\delta\|_2^2). \quad (5)$$

We compare $\hat{D}_{\text{Full}} = \frac{1}{2} \delta^\top \hat{F} \delta$ with $\hat{D}_{\text{Diag}} = \frac{1}{2} \sum_i \hat{F}_{ii} \delta_i^2$.

3.3 Routing-transition diagnostic

For downstream layer ℓ and token t , let q be baseline router logits and $\gamma = q_{(k)} - q_{(k+1)}$ the baseline top- k boundary gap. For modified logits q' , define

$$\rho_{\ell,t} = \frac{\|q'_{\ell,t} - q_{\ell,t}\|_{\infty}}{\gamma_{\ell,t}}, \quad \rho(x, \delta) = \max_{\ell,t} \rho_{\ell,t}. \quad (6)$$

A flip occurs when any monitored downstream top- k set changes. Because $\rho(x, \delta)$ is a maximum over a forward pass, it is a record-level risk summary, not an event-level calibrated probability or causal threshold.

4 Experimental protocol

Model and execution. We evaluate Qwen3-30B-A3B-Base, a 128-expert, top-8 MoE, at zero-indexed layer 23. The target-layer contribution dimension is 2,048. The model is split over two NVIDIA A100 80GB GPUs and run in strict FP32 with eager attention, TF32 disabled, and FP64 log-softmax/KL evaluation. Seeds, configs, versions, hashes, and memory statistics are stored with each run.

Data and target protocol. We use GSM8K (Cobbe et al., 2021), pinned to revision 740312add88f781978c0658806c59bc2815b9866. A seeded permutation of the official train split produces 512 build and 128 disjoint held-out prefixes after filtering to 31–127 tokens. Answers are excluded from primary manifests. The prompt asks the model to solve the given problem and provide the answer. To avoid fixed-suffix routing collapse, the intervention is at the full-prompt token containing the question’s final non-punctuation content character; the evaluated distribution is the next token after the final prompt colon. The F build uses 512 prefixes; joint interventions use a fixed first 16-prefix held-out subset.

Response construction and interventions. For each build prefix, we sample $y \sim p_0(\cdot | x)$ with a fixed seed, compute $s(x, y)$, and accumulate $ss^{\top}$. The primary estimate activates 85 experts at least once; 52 are activated at least 10 times. Joint interventions sample only from these 52 coordinates. We cross $|S| \in \{1, 2, 4, 8\}$ with $\epsilon \in \{0.01, 0.03, 0.05\}$, drawing 10 interventions per cell (120 total). The primary local analysis is $\epsilon = 0.01$: 40 interventions and 640 intervention–prefix records. The full grid gives 1,920 records for routing analysis.

Metrics and uncertainty. At intervention level we report R^2 , Pearson r , Spearman ρ_s , MSE, and MAE. We use 20,000 paired bootstrap resamples of the 40 local interventions for full-minus-diagonal comparisons. Routing groups use one-sided Mann–Whitney tests. Mean log-KL contrast and overall flip probability use 20,000 prefix-cluster bootstrap replicates. Wilson intervals on binned routing curves are descriptive and do not account for clustering.

5 Results

5.1 Implementation and estimator checks

At $\delta = 0$, logits are bit-identical to baseline; inactive-expert scores and model-parameter gradient counts are zero; target routing is unchanged; repeated scores agree exactly. A formal 10-prefix finite-difference validation at layer 47 tests 80 active and 20 inactive cases at $\epsilon = 0.01$. Active-case median relative error is 0.735%, the 90th percentile is 4.942%, and signs agree in 100% of cases. Layer 47 minimizes accumulated downstream roundoff; the same intervention adapter and algebra are used at primary layer 23.

The $N = 512$ matrix is symmetric to machine precision, agrees with offline $S^{\top}S/N$ to relative error 1.57×10^{-16} , and is positive semidefinite up to $\lambda_{\min} = -2.24 \times 10^{-20}$. Its trace is 4.17×10^{-4} , numerical rank is 80, and effective rank is 8.32. Figure 1 shows its diagonal and cross-response structure.

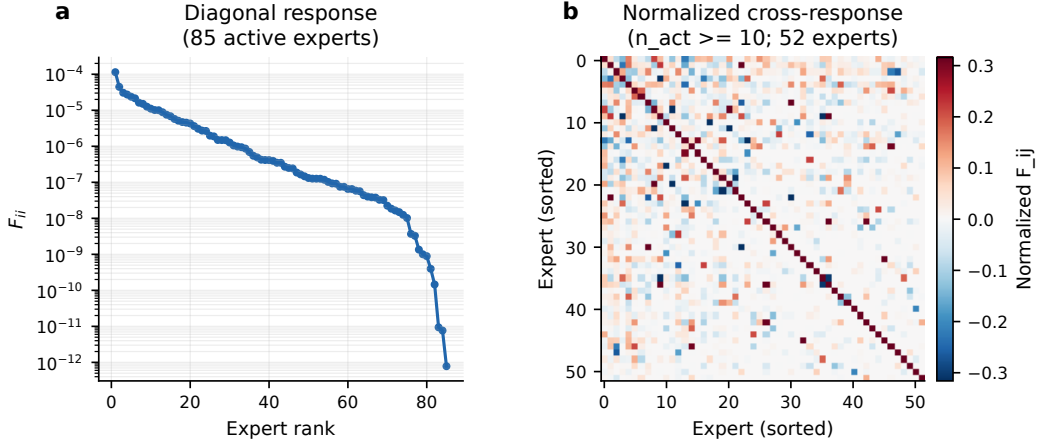


Figure 1: Estimated response geometry from 512 build prefixes. Left: F_{ii} for 85 experts activated at least once. Right: normalized signed cross-response for 52 experts activated at least 10 times. Coordinates outside observed routing support are not interpreted as zero functional response.

Local joint-intervention prediction (epsilon=0.01; no routing flips)

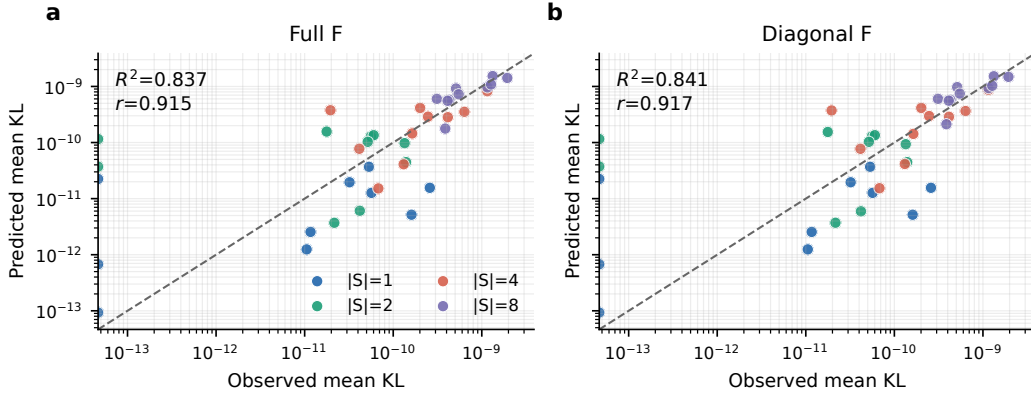


Figure 2: Local joint-intervention prediction at $\epsilon = 0.01$. Each point is observed or predicted next-token KL averaged over 16 held-out prefixes; colors indicate perturbed expert count. All 40 interventions preserve routing. Plotting zeros are placed below the smallest positive value only for log-scale display; statistics use unmodified values. Dashed line: identity.

5.2 Local effects are predictable; off-diagonal gain is absent

All 640 local prefix records preserve downstream routing. The full response achieves $R^2 = 0.837$, Pearson $r = 0.915$, and Spearman $\rho_s = 0.783$ (Figure 2, Table 1), supporting local quadratic prediction of unseen joint perturbations. The diagonal baseline is equally accurate: $R^2 = 0.841$, $r = 0.917$, and $\rho_s = 0.782$. A paired bootstrap gives $\Delta R^2 = -0.0040$ with 95% CI $[-0.0206, 0.0292]$ and bootstrap probability 0.379 that full R^2 exceeds diagonal. Thus we do not support the stronger hypothesis that estimated off-diagonal terms add predictive information.

For $|S| = 2, 4, 8$, median relative full-diagonal prediction differences are only 0.19%, 1.20%, and 3.14%. A matched high-interaction/control design appeared promising on 32 prefixes

Table 1: Local prediction at $\epsilon = 0.01$ (40 intervention-level means).

Predictor	R^2	Pearson r	Spearman ρ_s	MSE	MAE
Full F	0.837	0.915	0.783	3.239×10^{-20}	1.307×10^{-10}
Diagonal F	0.841	0.917	0.782	3.161×10^{-20}	1.305×10^{-10}

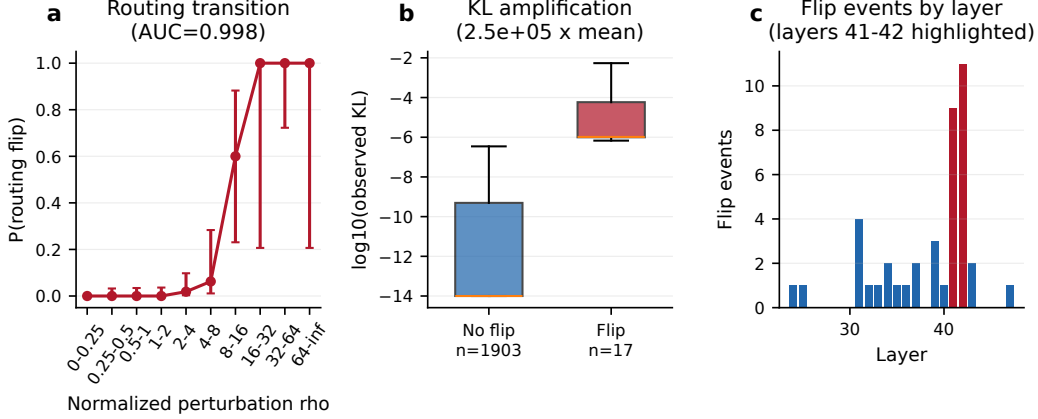


Figure 3: Routing transition and KL impact over 1,920 records. (a) Flip probability by prespecified bins of maximum normalized router displacement; bars show Wilson 95% intervals. (b) $\log_{10}(\text{KL} + 10^{-14})$ distributions with display outliers hidden. (c) Layers appearing in changed records; layers 41–42 are highlighted. These are record-level summaries.

but failed to replicate on all 128 held-out prefixes. We do not retune groups on those outcomes or claim confirmation.

5.3 Routing flips are rare, predictable, and high impact

Across the full grid, 17 of 1,920 prefix records contain a downstream flip, comprising 41 layer/token changes and affecting 14 of 120 interventions. The record-level ρ statistic sharply identifies this transition (Figure 3a): ROC-AUC is 0.9977 and flip ρ is larger by a one-sided Mann-Whitney test ($p = 4.24 \times 10^{-16}$). No flip occurs among 1,833 records with $\rho < 2$; all 12 with $\rho \geq 16$ flip. These are descriptive ranges, not selected thresholds.

Mean KL is 6.82×10^{-4} for flip records and 2.76×10^{-9} for no-flip records, a ratio of 2.47×10^5 ; one-sided Mann-Whitney $p = 3.15 \times 10^{-16}$. A prefix-cluster bootstrap gives 95% CI [6.77, 9.62] for the difference in mean $\log_{10}(\text{KL} + 10^{-15})$. We emphasize distributions and the cluster interval because no-flip values contain many exact zeros and groups are highly imbalanced. Flips concentrate late: 9 and 11 changed-record occurrences involve layers 41 and 42.

Removing all 14 flip-affected interventions restores Pearson $r = 0.815$ for full F , but R^2 remains slightly negative because this broader no-flip set contains scales outside the validated local regime. No discrete flip is thus necessary but not sufficient for second-order truncation accuracy.

5.4 Estimator instability limits geometry claims

Nested estimates converge slowly: relative Frobenius errors to F_{512} are 0.856 at $N = 128$, 0.453 at $N = 256$, and 0.237 at $N = 384$. Two disjoint 256-prefix halves differ by 0.827, with top-8 subspace overlap 0.535. Four categorical token samples per prefix change the matrix by 0.353 relative to one sample, yet its two halves still differ by 0.762.

On a new redesign split, categorical Rao–Blackwell and dense-tail Hutchinson estimators improve probe convergence but leave prefix-half differences of 0.566–0.953. Build-only shrinkage selected $\alpha = 0.25$, but on an 11-intervention no-flip tuning smoke its MSE (4.177×10^{-19}) is slightly worse than raw (4.164×10^{-19}). We stopped this redesign and did not open the independent confirmation set (Appendix D).

6 Discussion

Within a small routing-stable neighborhood, the score second moment predicts held-out joint effects well. This validates an intervention-space Taylor approximation and makes functional sensitivity tractable without 128 backward passes. The result is narrower than a stable global expert geometry.

Cross-expert terms do not improve prediction in the present grid, and targeted matched groups do not replicate. This can reflect weak interactions under the tested perturbation distribution, high off-diagonal estimation variance, or both; observed prefix heterogeneity prevents a purely structural conclusion.

Routing transitions explain large-error events. The strong ρ separation suggests that continuous response studies should report router-margin diagnostics and stratify discrete transitions. It does not make ρ a universal threshold because the statistic maximizes over correlated locations.

7 Limitations

We study one checkpoint, one primary layer, one task distribution, and next-token rather than sequence behavior. The local result has 40 interventions on 16 prefixes. E3 reuses these runs and is explanatory rather than independent replication; 17 flip records give limited transition-region resolution. The estimate is conditional on GSM8K routing support: 43 experts are never selected in the build sample. Post-training checkpoint comparisons and behavior-level reasoning evaluations have not passed the estimator-stability gate, so we make no claims about them.

8 Conclusion

Post-gating expert contributions admit a useful local response geometry: a single-backward score matrix predicts held-out joint effects when perturbations are small and routing remains stable. Downstream routing transitions are rare but dominate large KL changes, and a normalized margin statistic identifies them sharply. Full F does not outperform diagonal sensitivity, and matrix geometry remains prefix-unstable. Continuous response, discrete routing, and estimator uncertainty should be evaluated as separate but coupled components.

AI use statement

Generative AI tools assisted experimental code development, debugging, workflow documentation, analysis scripts, figure and table generation, and drafting and editing this manuscript. The authors reviewed AI-assisted outputs against saved configs and artifacts, executed numerical consistency checks, and retained negative results. No generative AI system is an author. The human authors take responsibility for the final content, claims, code, and artifacts. This disclosure must be updated before submission if subsequent uses change its scope.

Reproducibility statement

Supplementary code and artifacts specify checkpoint, pinned dataset revision, prompt and target-token protocol, split hashes, intervention definition, precision, seeds, and acceptance checks. One script regenerates every figure and result table from saved records. Appendices

report implementation validation and estimator uncertainty. The reserved confirmation set is excluded.

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A Implementation validation

Table 2: Numerical and algebraic validation checks; all listed checks pass.

Check	Setting	Result
Zero intervention	layer 23, FP32	max logit difference 0
Inactive experts	layer 47, 20 cases	autograd and FD max 0
Finite difference	layer 47, 80 active	median error 0.735%
Finite difference	layer 47, 80 active	P90 error 4.942%
Finite difference	layer 47, 80 active	sign agreement 100%
Repeat score	fixed seed/prefix	max difference 0
F symmetry	$N = 512$	relative error 0
Streaming vs. offline	$N = 512$	1.57×10^{-16}
Positive semidefinite	$N = 512$	$\lambda_{\min} = -2.24 \times 10^{-20}$

Layer 47 was used for the formal finite-difference gate to minimize accumulated downstream roundoff; the same adapter and algebra are used at layer 23. An initial BF16 score path was rejected because its numerical discrepancy was too large; all reported response and intervention results use strict FP32.

B Detailed local prediction results

Negative R^2 for small subsets reflects tiny observed KL and only 10 cases per stratum. Full and diagonal are identical for $|S| = 1$ by construction.

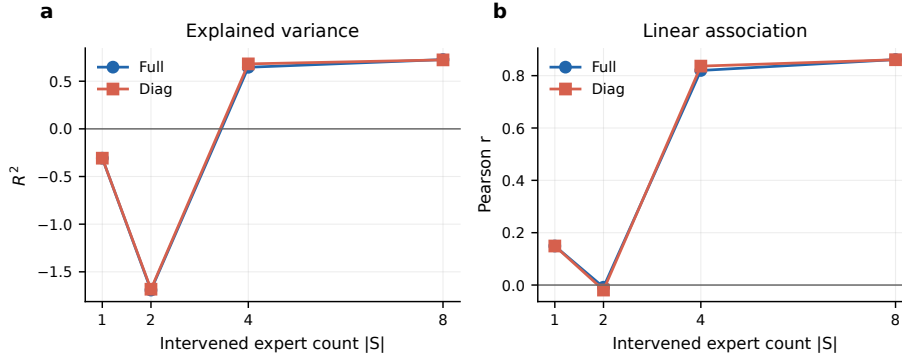


Figure 4: Descriptive local metrics by intervention subset size. Each stratum contains 10 interventions; the pooled result is primary.

Table 3: Local prediction stratified by intervened expert count.

$ S $	Predictor	n	R^2	Pearson r	Spearman ρ_s
1	Full	10	-0.308	0.149	0.423
1	Diagonal	10	-0.308	0.149	0.423
2	Full	10	-1.691	-0.008	0.067
2	Diagonal	10	-1.682	-0.019	0.067
4	Full	10	0.646	0.820	0.503
4	Diagonal	10	0.682	0.836	0.503
8	Full	10	0.727	0.861	0.903
8	Diagonal	10	0.724	0.861	0.879

C Routing analysis details

D Sampling stability and estimator redesign

E Artifact scope

All figures are generated from saved JSONL, JSON, CSV, and NumPy artifacts. The local prediction unit is an intervention mean over 16 prefixes; the routing unit is one intervention-prefix record. These units remain separate to avoid inflating sample size. The generation manifest records output hashes and that the redesign confirmation set was not accessed.

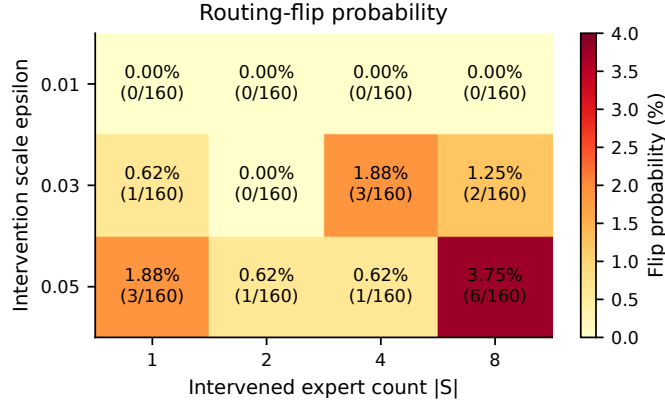


Figure 5: Flip probability by perturbation scale and subset size. Each cell contains 160 records. No record flips at $\epsilon = 0.01$. Percentages are descriptive because records share prefixes and interventions.

Table 4: Record-level routing-transition summary.

Quantity	Estimate	Uncertainty/reference
Records	1,920	16 prefixes $\times$ 120 interventions
Records with flip	17	cluster rate CI [0.0021, 0.0203]
Affected interventions	14	of 120
ρ ROC-AUC	0.9977	Mann-Whitney $p = 4.24 \times 10^{-16}$
Median ρ , flip	33.415	no-flip P90 = 0.892
Mean KL, flip	6.822×10^{-4}	no-flip = 2.765×10^{-9}
Mean log-KL contrast	flip minus no-flip	cluster CI [6.77, 9.62]

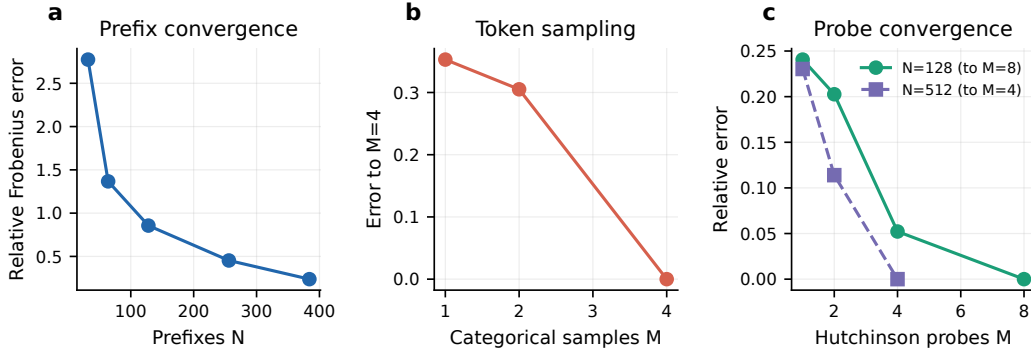


Figure 6: Sampling diagnostics: nested-prefix estimates, categorical token sampling, and Hutchinson probe convergence. Probe convergence does not imply convergence over independent prefixes.

Table 5: Prefix-split stability; lower difference and higher overlap are better.

Estimator	N	Half difference	Top-8 overlap	Decision
Categorical $M = 4$	512	0.762	0.613	unstable
Categorical RB $M = 4$	128	0.953	0.693	unstable
Hutchinson $M = 8$	128	0.892	0.564	unstable
Hutchinson $M = 4$	512	0.566	0.631	narrow gate fail

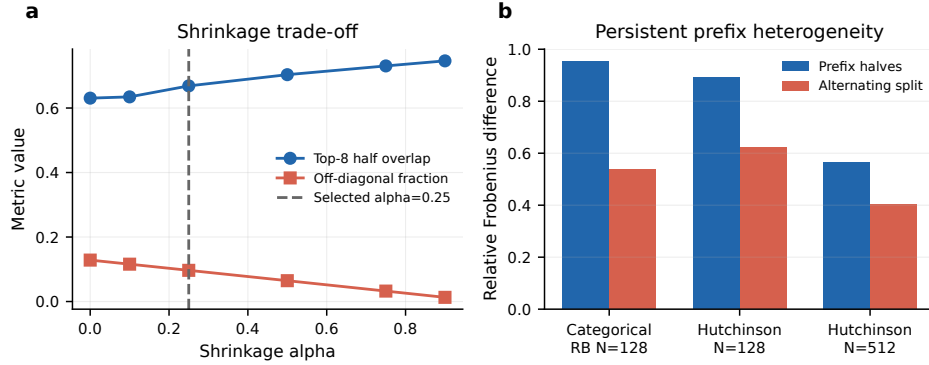


Figure 7: Exploratory redesign. Shrinkage trades subspace stability against off-diagonal magnitude, while prefix heterogeneity persists across estimators. Outcome tuning did not select a superior estimator, so confirmation stayed sealed.

Table 6: Exploratory no-flip shrinkage tuning smoke ($n = 11$).

Predictor	R^2	MSE	MAE
Shrinkage ($\alpha = 0.25$)	0.5042	4.1769×10^{-19}	4.5973×10^{-10}
Raw response	0.5057	4.1644×10^{-19}	4.5792×10^{-10}
Diagonal	0.4991	4.2201×10^{-19}	4.6515×10^{-10}